

Erdős Problem #530

EP-530

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\ell(N)$ be maximal such that in any finite set $A \subset \mathbb{R}$ of size N there exists a Sidon subset S of size $\ell(N)$ (i.e. the only solutions to $a + b = c + d$ in S are the trivial ones). Determine the order of $\ell(N)$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>